

Examples

Iterative Time Complexity

Disclaimer : These examples are from popular internet sources!

```
for (int i = 0; i < n; i++) {  
    // regular statements running in  
    // constant time  
}
```

```
for i in range(0, n):  
    // regular statements running in  
    // constant time
```

$O(n)$

```
for (int i = n; i > 0; i--) {  
    // regular statements running in constant time  
}
```

$O(n)$

```
for (int i = 0; i < n; i++) {  
    for (int j = 0; j < n; j++) {  
        // regular statements running in constant time  
    }  
}
```

$O(n^2)$

```
for (int i = 0; i < n; i++) {  
    for (int j = 0; j < i; j++) {  
        // regular statements running in constant time  
    }  
}
```

$O(n^2)$

```
for (int i = 0; i < n; i++) {  
    // regular statement running in constant time  
}
```

```
for (int j = 0; j < n; j++) {  
    // regular statement running in constant time  
}
```

$O(n)$

```
q=0;  
for (int i = 1; q <= n; i++) {  
    q=q+i;  
}
```

$O(\sqrt{n})$

```
for (int i = 1; i < n; i=i*2) {  
    // regular statements running in constant time  
}
```

Hint : $\log_a x = (\log_b x) / (\log_b a)$

$O(\log_2 n)$

```
for (int i = n; i >= 1; i=i/2) {  
    // regular statement running in constant time  
}
```

Hint : $\log_a x = (\log_b x) / (\log_b a)$

$O(\log_2 n)$

```
q=0;  
for (int i = 1; i < n; i=i*2) {  
    q++;  
}
```

```
for (int j = 1; j < q; j=j*2) {  
    // regular statement running in constant time  
}
```

$O(\log \log_2 n)$

```
for (i=n, i>=1; i=i/3){  
    p=2;  
    while(p<n){  
        p=p*2;  
    }  
}
```

$O(\log_3 n * \log_2 n)$

```
for (i=n/2, i<= n; i++){  
    for (j=1; j+n/2 <=n; j++){  
        for (k=1; k <=n; k = k*2){  
            // regular statement running in constant time  
        }  
    }  
}
```

$O(n^2 \cdot \log_2 n)$